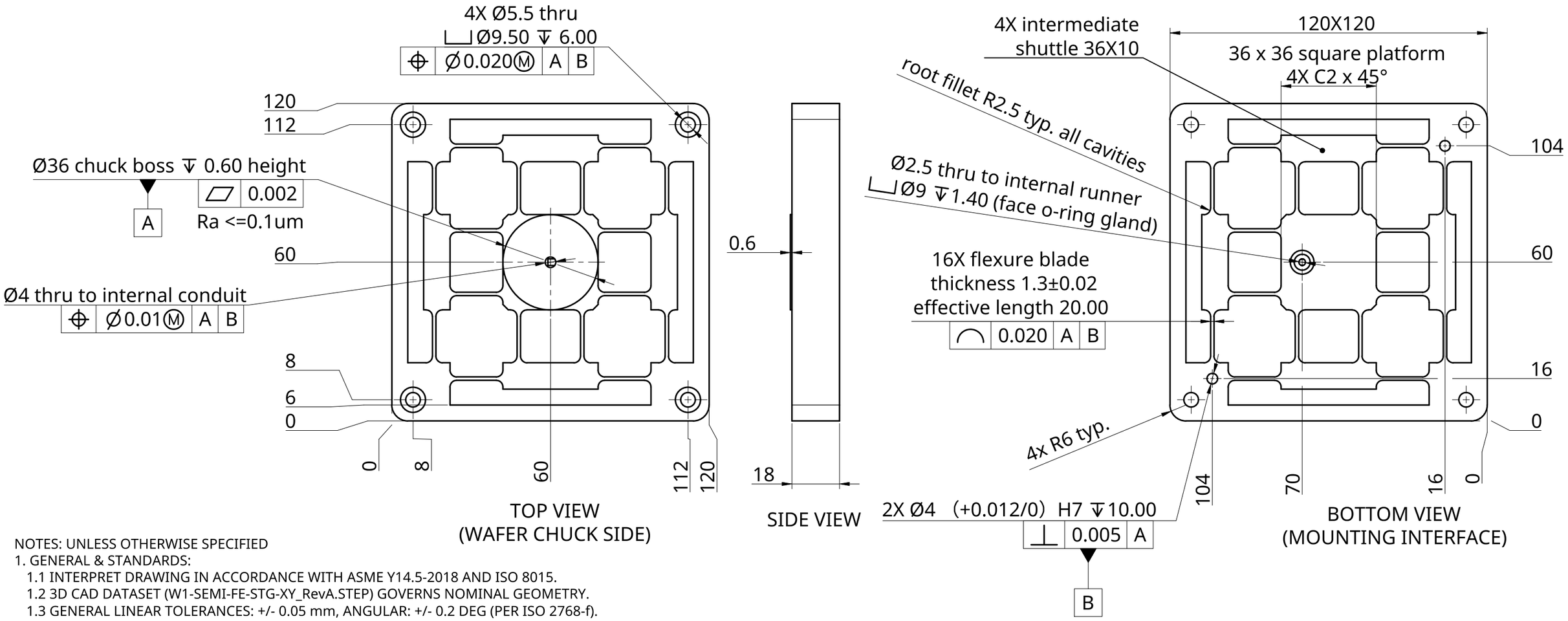


NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W1-SEMI-FE-STG-XY_RevA.STEP) GOVERNS NOMINAL GEOMETRY.
   1.3 GENERAL LINEAR TOLERANCES: +/- 0.05 mm, ANGULAR: +/- 0.2 DEG (PER ISO 2768-f).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & THERMAL POST-PROCESSING:
   2.1 PRIMARY MATERIAL: ADDITIVE GRADE Ti-6Al-4V ELI (GRADE 23) PER ASTM F136.
   2.2 HOT ISOSTATIC PRESSING (HIP): 920°C +/- 10°C AT 120 MPa FOR 180 MINS UNDER ARGON.
   2.3 MICROSTRUCTURE: HOMOGENEOUS AND DENSE ALPHA+BETA DUPLEX NEEDLE MARTENSITE; DENSITY > 99.98%.
   2.4 THEORETICAL CALCULATED NET MASS: 312.0 g (NET VOLUME: 70,428 mm^3).

3. SURFACE ENGINEERING & MACHINING CONTROLS:
   3.1 WAFER CHUCK CONTACT SURFACE (DATUM A) FLATNESS <= 0.002 mm; SURFACE FINISH Ra <= 0.1 MICRONS.
   3.2 FLEXURE BLADES (16X) WALL THICKNESS 1.30 +/- 0.02 mm WITH FULL R2.5 mm ROOT BLENDS.
   3.3 FINISH FLEXURE CAVITIES VIA WIRE EDM OR HIGH-SPEED MICRO-MILLING (NO HEAT MARKS/MICROCRACKS).
   3.4 INTERNAL CONDUIT ISOLATION WALL THICKNESS MUST EXCEED 1.25 mm THROUGHOUT.

4. ULTRA-HIGH VACUUM & CLEANLINESS COMPLIANCE:
   4.1 COMPONENT COMPLIES WITH SEMI E49 / SEMI E57 ULTRA-HIGH VACUUM OUTGASSING LIMITS.
   4.2 NO HYDROCARBONS, ADHESIVES, ELASTOMERS, OR EMBEDDED SLAG RESIDUES PERMITTED.
   4.3 VACUUM LEAK RATE: GLOBAL HELIUM LEAK RATE < 1.0E-10 mbar*L/s PER ASTM E498.

5. CLEANROOM HANDLING & PACKAGING:
   5.1 ULTRASONIC DEGREASING IN REAGENT-GRADE IPA FOLLOWED BY HIGH-PURITY DI WATER RINSE.
   5.2 INSPECTED AND PACKAGED IN ISO CLASS 3 (FED-STD-209E CLASS 1) CLEANROOM ENVIRONMENT.
   5.3 DOUBLE-BAG SEALED UNDER PURGED HIGH-PURITY NITROGEN (N2) IN ANTISTATIC PE ENVELOPE.
   5.4 EACH PACKAGE MUST BE PERMANENTLY LABELED WITH DWG NO, LOT NO, AND CLEANROOM DATE.

		NAME	SIGNATURE	DATE
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS	DRAWN			2026-10-14
ANGULAR = ± °	CHECKED			
SURFACE FINISH	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				

THIRD ANGLE PROJECTION

MATERIAL	FINISH
Ti-6Al-4V ELI (Grade 23) per ASTM F136	

TITLE: DUAL-AXIS DECOUPLED COMPLIANT WAFER CHUCK STAGE

SIZE	DWG NO.	REV.
A3	W1-SEMI-FE-STG-XY_REVA	A

SCALE	WEIGHT	SHEET
NTS	0.312 kg	1 of 1